

Thermo-Viscoelastic Laminate-Based Finite Element Modeling of Fused Filament Fabrication Direct 4D Printing

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A. Supplementary Information – Prony Series Characterization

A.1 Material Model and Data Processing

The shifting procedure used to determine C_3 is conducted using a numerical optimization scheme based on a Nonuniform Pattern Search (NUPS) algorithm in MATLABs inbuilt function `patternsearch`. The objective is to shift the individual frequency sweeps $E'_{exp}(f)$ to create a master curve that is as smooth as possible by minimizing the absolute second derivative along its entire length (second column Fig. 5 in the main text). This is achieved by fitting a spline through the individual data points gathered during the mechanical testing procedure and numerically determining the second derivative as shown in Equation (S1).

$$\min_{C_3} \sum_i \int_{f_{min}}^{f_{max}} \left| \frac{d^2 E'_{exp}(C_3, f)}{df^2} \right| \quad (S1)$$

$$s.t. \quad 0 < C_3 \leq Inf$$

A Generalized Maxwell Model consisting of $n = 23$ Maxwell elements is fitted to the experimental data. The fitting procedure is conducted in the frequency domain using a numerical optimization scheme based on a nonlinear least-squares algorithm (Equation (S2)). To improve performance and convergence during optimization, the optimizer sets the coefficient τ_n and the exponent τexp_n of the individual relaxation times separately.

$$\min_{k_n, \tau_n, \tau exp_n} \sum_{\log f} \left(E'_{exp}(\log f) - \left(k_0 + \sum_n \frac{k_n \cdot (10^{\log f})^2 \cdot (\tau_n \cdot 10^{\tau exp_n})^2}{1 + (10^{\log f})^2 \cdot (\tau_n \cdot 10^{\tau exp_n})^2} \right) \right)^2 \quad (S2)$$

$$s.t. \quad 1 \leq k_n \leq Inf$$

$$1 \leq \tau_n \leq 10$$

$$min(\log f) - 5 \leq \tau exp_n \leq max(\log f) + 5$$

The fitted models are expressed using a normalized Prony series. The stiffness of the Maxwell elements is thus defined as a fraction of the long-term stiffness k_0 (third column Fig. 5 in the main text).

A.2 AM Pre-Strain Determination

The AM pre-strain is determined by an optimizer that controls the pre-strain in the active material of the bilayer and evaluates the absolute difference in curvature between the AM and simulated sample to find the pre-strain that creates the desired curvature as shown in Equation (S3):

$$\min_{\varepsilon} \left(\kappa_{exp} - \kappa_{sim}(\varepsilon) \right)^2 \quad (S3)$$

$$s.t. \quad 0.01 \leq \varepsilon \leq 0.15$$

The experimental curvature κ_{exp} is determined by the manufactured bilayers. The simulated curvature $\kappa_{sim}(\varepsilon)$ is determined from the final state of the FE simulation. The bilayers are simulated in Abaqus/Standard (Dassault Systemes, Vélizy-Villacoublay, France) using four node quadrilateral shell elements (S4R). The input file for the simulation is generated in MATLAB, the fitting procedure is conducted using a numerical optimization scheme based on a Nonuniform Pattern Search (NUPS) algorithm in MATLABs inbuilt function *patternsearch*. This specific algorithm is chosen since the finite element simulation acts as a black box objective function. During the AM pre-strain determination, the optimization method matches the curvature of the benchmark samples, thus the error is given by the convergence criteria of the optimization method. The optimization uses the default value provided by MATLAB for the “FunctionTolerance”, which is equal to 10^{-5} .

B. Supplementary Information – Results

B.1 Prony Series Characterization

Long-term mechanical properties of both PLA and PETG are shown in Table B.1. The resulting elastic element stiffness, k_i , and the damping relaxation time τ_i of the individual Maxwell elements are shown in Table B.2.

Table B.1: Long-term properties of both PLA and PETG. Properties include the long-term Young’s modulus in the rubbery state k_0 , the Poisson coefficient ν , the glass transition temperature T_g , and the constants required by the WLF and Arrhenius equation C_1 - C_3 .

Parameter		PLA	PETG
k_0	[MPa]	10.598	8.093
ν	[–]	0.35	0.38
T_g	[°C]	65	85
C_1	[–]	17.4	17.4
C_2	[–]	51.6	51.6
C_3	[–]	35000	65283
α	[$\mu\text{m}/\text{mK}$]	68.0	68.4

Table B.2: Prony series parameters for all 23 Maxwell elements of both PLA and PETG.

Parameter	PLA		PETG	
	k_i [MPa]	τ_i [s]	k_i [MPa]	τ_i [s]
1	6.228	1.176E-14	19.582	5.228E-25
2	1.806	1.228E-14	11.809	1.945E-22
3	7.849	2.878E-14	12.217	5.347E-20
4	12.751	1.701E-12	15.028	3.018E-18
5	20.857	9.178E-12	14.473	1.058E-16
6	31.104	6.034E-11	15.538	3.149E-15
7	45.566	6.187E-09	19.468	6.582E-14
8	64.036	1.102E-07	24.738	1.006E-12
9	92.484	1.047E-06	31.328	1.200E-11
10	135.916	6.903E-06	42.325	1.141E-10
11	113.488	2.582E-05	56.216	8.740E-10
12	128.266	5.363E-05	77.873	5.834E-09
13	173.470	1.838E-04	101.924	3.203E-08
14	93.393	4.386E-04	114.156	1.636E-07
15	163.804	7.577E-04	117.252	8.021E-07
16	134.311	1.795E-03	114.524	3.829E-06
17	119.266	2.292E-03	103.643	1.830E-05
18	100.596	6.576E-03	91.709	8.782E-05
19	80.534	9.920E-03	79.062	4.259E-04
20	61.624	3.290E-02	49.856	2.062E-03
21	44.508	5.374E-02	18.940	1.212E-02
22	30.529	6.586E-02	3.549	3.074E-01
23	18.610	3.503E+01	1.797	1.675E+01

23 Maxwell elements are chosen to ensure a trade-off between model accuracy and model size. An analysis of the quality of the fitting parameters is shown in Fig. B.1. A higher number of Maxwell elements is chosen to better approximate the high-frequency behavior of both curves. Especially for the PETG, a long slow increase of stiffness with frequency can be observed when the polymer is in its glassy state (see Fig. 5b). The generalized Maxwell model can be seen as an approximation of the master curve with a series of Sigmoid functions. While the root mean square error (RMSE) did not change significantly above 15 Maxwell elements, an inspection of the high frequency section (shown in Fig. B.1a) of the model showed better master curve approximation for slightly higher Maxwell element counts.

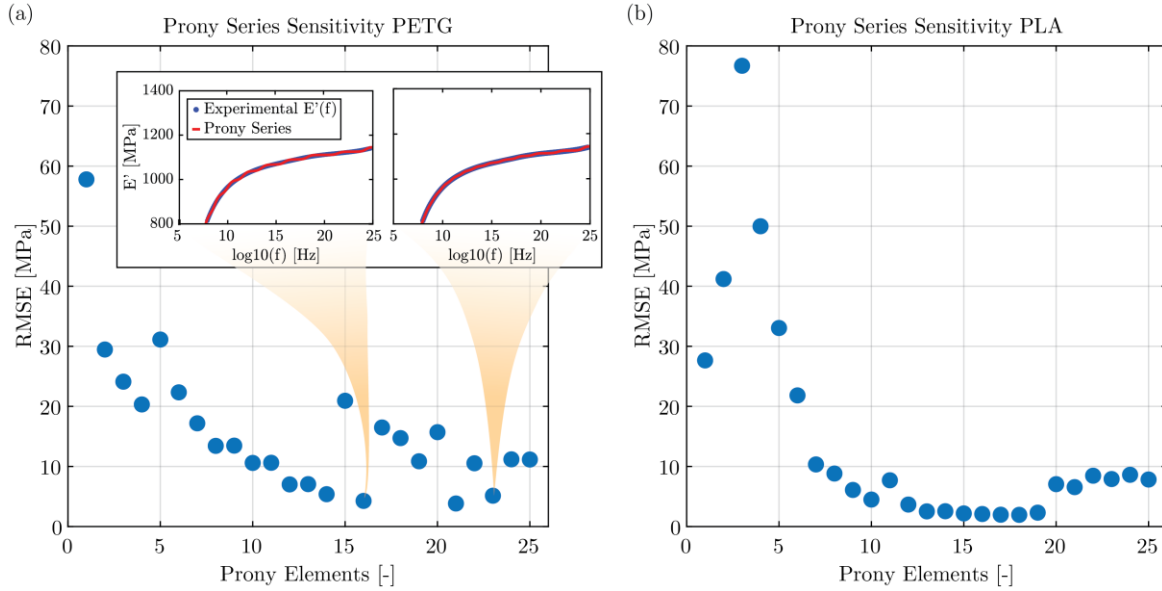


Fig. B.1. RMSE of the Prony series when compared to the experimental master curve for PETG (a) and PLA (b), evaluated for Maxwell element counts between one and 25. A direct comparison of the high frequency section for PETG is shown to illustrate how better fitting is observed at higher element counts even with only small changes in RMSE.

B.2 Basic Shape Change and Time-Temperature Dependent Behavior Stress-Strain Plots

The von Mises stress and logarithmic strain of the three sample objects from Section 3.2.1 are shown in Fig. B.2. In Fig. B.2a the boundary condition at the top of the braid causes a local stress reduction due to the pinned nodes. Two distinct strain zones can be observed as expected from the model definition in Section 2.3.1. The box in Fig. B.2b shows localized stress and strain in the hinges, the box faces remain inactive. The twisting tube shows local stress concentrations and elevated strains at the back edge. This is due to the fully fixed (*encastre*) boundary condition introduced at this back edge.

The von Mises stress and logarithmic strain of the sequential bilayer from Section 3.2.1 are shown in Fig. B.3. The von Mises stress results for the sequential bilayer are directly related to the current temperature of the simulation. The highest stresses are observed during the initial activation of the PLA at 65°C. This is due to the PETG still being in its glassy state and the PLA activating. The higher stiffnesses of the materials at this temperature (see Section 4.2.1) cause higher stresses. As the bilayer is heated further, the stiffnesses of both PLA and PETG drop significantly. At the maximum temperature of 100°C, both are being governed primarily by the stiffness of k_0 (see Table B.1) leading to much lower stress values. The logarithmic strain is directly related to the current activation state of the sequential bilayer.

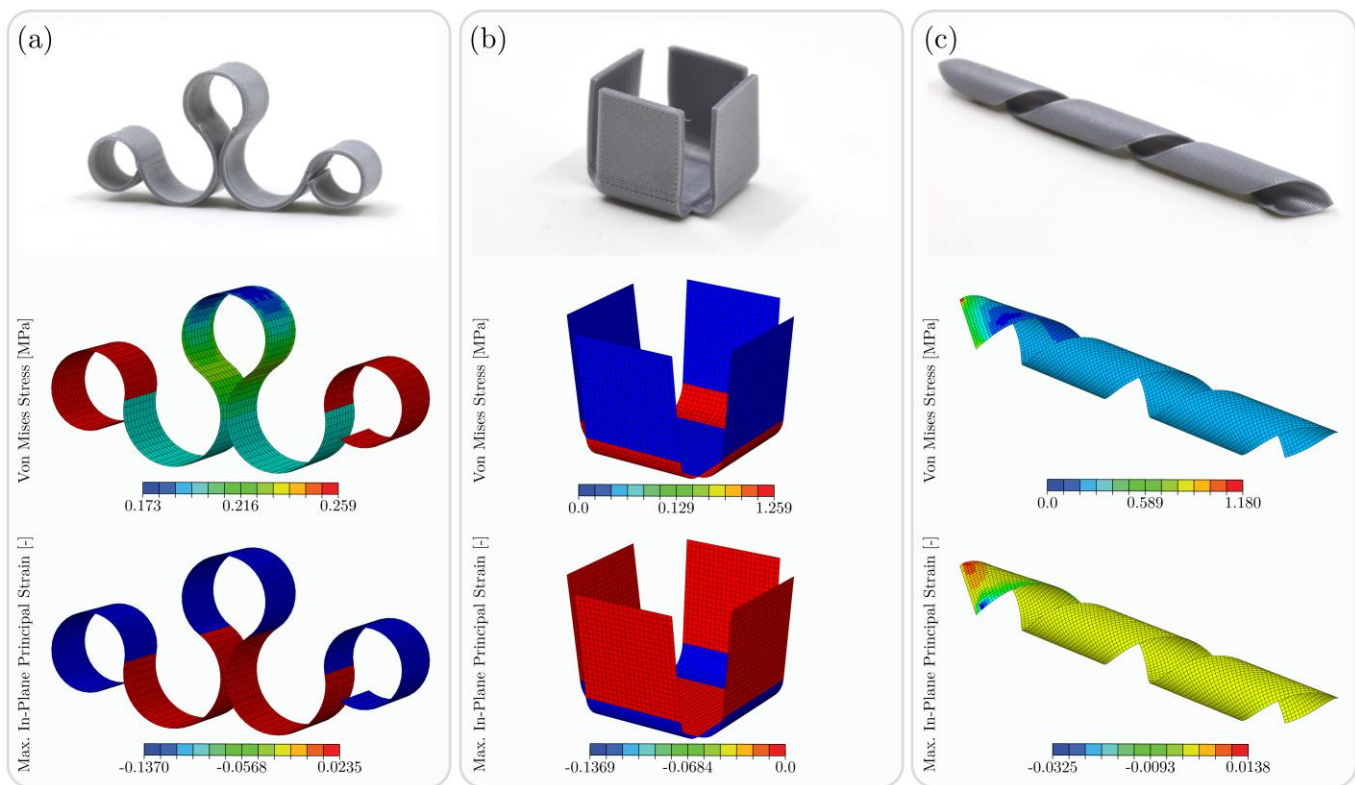


Fig. B.2. Fabricated activated states as well as the simulated von Mises stress distribution and logarithmic strain of the activated state of all three sample objects.

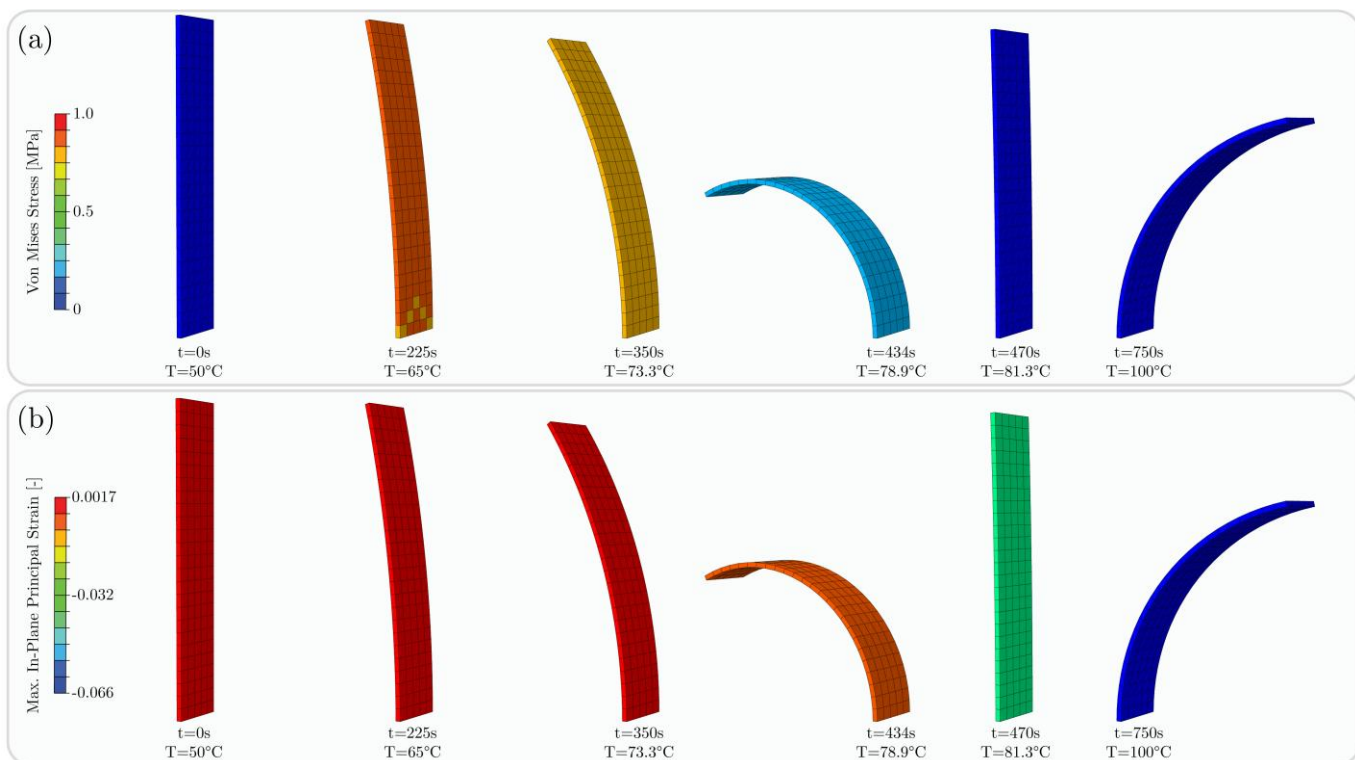


Fig. B.3. Von Mises stress (a) and logarithmic strain (b) distribution in the simulated sequentially activating PLA-PETG bilayer.

B.3 Additional Structures of Increased Complexity

A Miura-Ori origami [1] and a flower with twisting petals [2] are shown in Fig. B.4. These two additional examples illustrate how the modeling approach can be used in more complex structures. In structures, such as the origami pattern, that exhibit a global rigid mode, the model is able to accurately actuate this global mode at a hinge level and predict the deformed shape without numerical instabilities due to the mode. Both models are simulated using an active PLA section. The Miura-Ori pattern is additively manufactured using a nozzle temperature of 200°C and an activation temperature of 90°C. The twisting flower uses the same settings as the twisting sample from Fig. 6c.

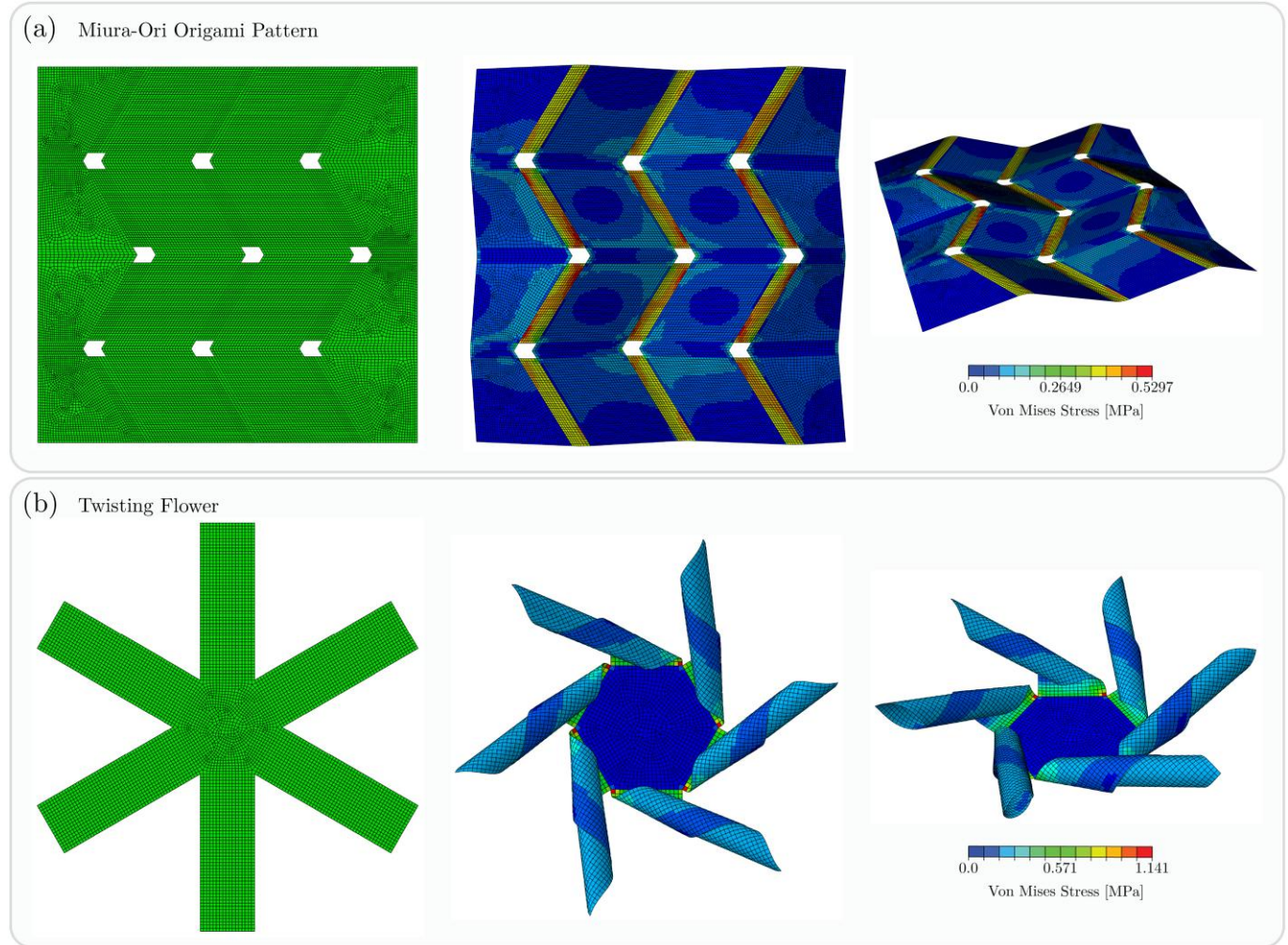


Fig. B.4. (a) Simulated von Mises stress and activated shape of a Miura-Ori origami pattern. (b) Simulated von Mises stress and activated shape of a twisting flower pattern.

B.4 Comparison to Linear and Nonlinear Thermal Expansion

A direct comparison to linear and nonlinear thermal expansion modeling approaches can be drawn based on Fig. B.5. In the case of linear thermal expansion, the AM pre-strain is applied over the full temperature range between 22°C and 95°C. Both PLA and PETG are still modeled as viscoelastic materials. The AM pre-strain for PLA is modeled as a thermal strain of $-576 \mu\text{m}/\text{mK}$, resulting in a total strain of -4.206% at 95°C as stated in Table 5 of the main text. Similarly, the AM pre-strain for PETG is modeled as a thermal strain of $-826 \mu\text{m}/\text{mK}$, resulting in a total strain of -6.029% at 95°C. As both materials are activating simultaneously in this modeling approach, the bilayer almost immediately starts displacing towards the PETG. While the final position of the bilayer at 26.7° is only 13.6% smaller than the experimental mean of 30.9° at 95°C, the path the bilayer takes to get there shows no match with the fabricated sequential bilayer.

In contrast, nonlinear thermal expansion requires much more significant tuning. For the current simulation approach based on Abaqus, nonlinear thermal expansion is temperature based. A series of thermal expansion coefficients and corresponding temperatures are provided to Abaqus. To model a shape memory effect, most of the AM pre-strain should be released above the glass transition temperature T_g . Abaqus interpolates between individual thermal expansion values, the final thermal strain is the integration of this interpolated thermal expansion coefficient over the simulated temperature range. Prior knowledge of how the AM pre-strains of the two base materials are released relative to each other must be known to accurately model the sequential bilayer. This circumstance mandates a fitting procedure of the thermal strain based on the specific fabricated sequential bilayer being simulated. This necessitates a case-specific fitting procedure for each fabricated bilayer. Although this method can replicate the observed bilayer behavior after fitting, it lacks predictive capability based solely on the material properties and AM pre-strain. In contrast, the modeling approach proposed in this work only requires knowledge of the base materials. Any material, geometry and layer combination can be simulated directly without the need for a case-specific fitting procedure, thus offering a significant advantage in terms of predictive accuracy and generalizability.

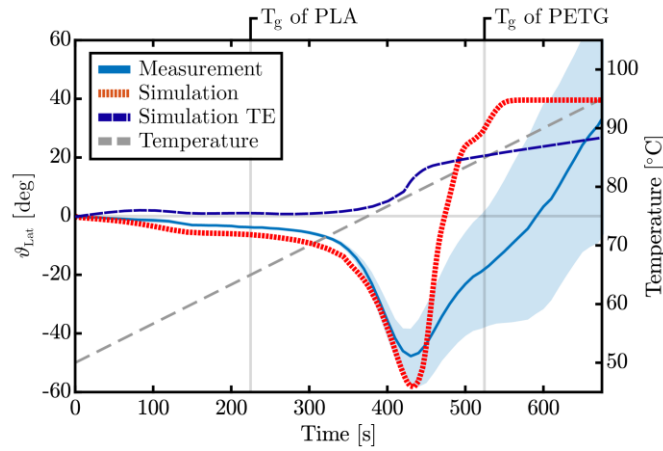


Fig. B.5. Time-temperature dependent angular displacement of the fabricated and simulated sequential bilayers using two simulation models. One simulation uses the modeling approach proposed in this work and one uses a linear thermal expansion approach.

C. Supplementary Information – Simulation Files

C.1 Sample Objects and Sequential Bilayer

All three sample object input files can be found in the directory “01_SampleObjects”. The full simulation for all three objects is included in the single “.inp” file. To run the simulations, they need to be submitted to Abaqus together with the user subroutine “UMAT_SMPViscoShells.f”.

- 01_SampleObjects \Bending_Braid.inp
- 01_SampleObjects \Bending_Box.inp
- 01_SampleObjects \Twisting_Tube.inp

The simulation files for the sequential bilayer can be found in the directory “02_SequentialBilayer”. The simulation input is split into two files, the “.inp” contains the simulation parameters and the “.inc” file contains all geometry and the mesh. To run the simulation, the “.inp” needs to be submitted to Abaqus together with the “.inc” file and the user subroutine “UMAT_SMPViscoShells.f”.

- 02_SequentialBilayer \AbaqusSequentialBilayerInput.inp
- 02_SequentialBilayer \AbaqusSequentialBilayerInput.inc

C.2 Wave Spring

The simulation files for the wave spring can be found in the directory “03_WaveSpring”. The simulation input is split into two files, the “.inp” contains the simulation parameters and the “.inc” file contains all geometry and the mesh. To run the simulation, the “.inp” needs to be submitted to Abaqus together with the “.inc” file and the user subroutine “UMAT_SMPViscoShells.f”.

- 03_WaveSpring\AbaqusWaveSpringInput.inp
- 03_WaveSpring\AbaqusWaveSpringInput.inc

D. Supplementary Information – Video Files

D.1 Video S1

Wave spring simulation results showing both the temperature distribution and observed displacement of the simulated wave spring. The results are drawn directly from the simulation files provided in Section C.2.

D.2 Video S2

Direct comparison of the activation of the fabricated and simulated wave spring. The simulation results are drawn directly from the simulation files provided in Section C.2.

D.3 Video S3

Direct comparison of the activation of the fabricated and simulated sequential bilayer. The simulation results are drawn directly from the simulation files provided in Section C.1.

Bibliography

- [1] R. Wang, C. Yuan, J. Cheng, X. He, H. Ye, B. Jian, H. Li, J. Bai, Q. Ge, Direct 4D printing of ceramics driven by hydrogel dehydration, *Nat Commun* 15 (2024) 758. <https://doi.org/10.1038/s41467-024-45039-y>.
- [2] Z. Wang, Z. Wang, Y. Zheng, Q. He, Y. Wang, S. Cai, Three-dimensional printing of functionally graded liquid crystal elastomer, *Sci Adv* 6 (2020). <https://doi.org/10.1126/sciadv.abc0034>.